

SPL Compiler

Type A completeness

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Directory Structure

Ensure your project has the following structure:

```
Input/  
├─ program1.txt  
└─ program2.txt  
Compiler.jar
```

- Place all your SPL source files in the `Input` directory.
- The compiler jar (`Compiler.jar`) should be in the same root directory as `Input/` .

Running the Compiler

Run the compiler using:

```
java -jar Compiler.jar
```

You will be presented with a list of available SPL files in the `Input` directory.

- Enter the **number** corresponding to the file you want to compile.

Output

- The generated BASIC code will be saved in:

```
Output/<InputFileName>/<InputFileName>.txt
```

- The Intermediatecode.html will be saved in:

```
Output/<InputFileName>/<InputFileName>.html
```

For example, compiling `program1.txt` will produce:

```
Output/program1/program1.txt  
Output/program1/program1.html
```

- The `Output` directory will be created automatically if it doesn't exist.
- You can then view the intermediate code in a browser and run the basic code in a basic compiler -> We used [Applesoft Basic](#) ,[C64 Basic emulator](#) and `bwbasic` (linux program) to test the output code.
- The Basic compilers we tested with can only use max 2 char long variables, so we implemented smart temp variable assignment and reuse so it never needs to use something like `t10`, which will end up being defined as `t1` in the compiler. User defined variables are also shortened like that, with scope checks to ensure reuse is safe.